

# Region Affinities

## Data Appendix

### *Source Data, Tables, and Reference Materials*

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*Companion documents: Summary · Technical Appendix · Methods Primer · Region Reference — Identity · Region Reference — Terroir*

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## D.1 Identity Dimension Key

The six identity dimensions are scored as integers in  $\{-2, -1, 0, +1, +2\}$ . Each dimension is a bipolar axis whose poles are described below; the score expresses where the region sits on that axis, with 0 representing balance or neutrality.

| Code | Dimension                 | Positive pole (+2)            | Negative pole (−2)              |
|------|---------------------------|-------------------------------|---------------------------------|
| D1   | Interiority ↔ Exteriority | Interior, place-defined       | Outward-facing, market-oriented |
| D2   | Struggle ↔ Ease           | Extreme difficulty, endurance | Pleasure, comfort, ease         |
| D3   | Tradition ↔ Reinvention   | Deep custodial tradition      | Radical reinvention, disruption |
| D4   | Individual ↔ Collective   | Strongly individual, solitary | Strongly collective, communal   |
| D5   | Urgency ↔ Timelessness    | Urgent, present-focused       | Timeless, eternal               |
| D6   | Earthly ↔ Transcendent    | Deeply grounded, earthly      | Transcendent, spiritual         |

D-scores were assigned by the author based on the corresponding identity narrative for each region, then reviewed by a panel of subject-matter experts. Sixteen score adjustments were made across thirteen regions following expert review, documented in Section D.4.

## D.2 Identity Cluster Summary

Six identity clusters were detected by K-means on the standardised D-score matrix.

| Cluster               | Size | Defining character                                         |
|-----------------------|------|------------------------------------------------------------|
| The Moderates         | 15   | Balanced across all dimensions; equilibrium over extremity |
| New World Reinvention | 12   | Constructed identities, urgency, disruption                |
| Old World Interior    | 10   | Tradition-bound, place-defined, custodial                  |
| Against the Odds      | 10   | Defined by struggle, extreme terrain, marginalisation      |
| Old World Exterior    | 7    | Institutional, market-facing, tradition-holding            |
| Outward Ease          | 5    | Outward-facing, pleasure-oriented, accessible              |

## D.3 Terroir Cluster Summary

Seven terroir clusters were detected by K-means on the 10-component PCA projection of the TF-IDF terroir matrix. Cluster count, membership, and labels are fully model-derived.

**Mediterranean & Volcanic** (16 regions). Campania, Dalmatian Coast, Etna, Friuli-Venezia Giulia, Goriška Brda, Liguria, Macedonia, Piedmont, Santorini, Sardinia, Sicily, Steiermark, Swartland, Tokaj, Tuscany, Veneto.

**Southern Hemisphere & International** (12 regions). Alto Adige, Barossa Valley, Central Otago, Columbia Valley, Finger Lakes, Hawke’s Bay, Hunter Valley, Maipo Valley, Margaret River, Marlborough, Patagonia, Stellenbosch.

**French Viticultural** (10 regions). Alsace, Beaujolais, Bordeaux, Burgundy, Champagne, Châteauneuf-du-Pape, Jura, Loire, Northern Rhône, Provence.

**American West Coast** (7 regions). Napa Valley, Paso Robles, Santa Barbara, Santa Cruz Mountains, Sonoma, Walla Walla, Willamette Valley.

**Iberian Continental** (6 regions). Catalonia, Douro, Galicia, Mendoza, Ribera del Duero, Rioja.

**Germanic Rhine** (5 regions). Baden, Mosel, Nahe, Pfalz, Rheingau.

**Austrian Danube** (3 regions). Kamptal, Wachau, Wagram.

## D.4 SME Change Log

The following adjustments to D-scores and metaphors were made after subject-matter expert review of the initial scoring pass. Three of these changes resulted in cluster reassignments, noted in the rationale column.

| Region          | Metaphor      | Change                                  | Rationale                                                                  |
|-----------------|---------------|-----------------------------------------|----------------------------------------------------------------------------|
| Bordeaux        | Business      | D3: +1 → +2                             | Deeper institutional tradition; separates from Outward Ease                |
| Champagne       | Society       | D3: +1 → +2                             | Tradition as dominant institutional force                                  |
| Etna            | Awakening     | D1: −1 → +1, D2: +1 → +2, D3: −1 → 0    | Volcanic introspection is interior; cluster shift to Old World Interior    |
| Goriška Brda    | Fortune       | Metaphor change, D3: −1 → 0             | Resolves duplicate; accidental fortune, not reinvention                    |
| Rioja           | Patience      | Metaphor change                         | Resolves duplicate with Campania; captures extended-aging philosophy       |
| Santa Barbara   | Serendipity   | D3: −1 → 0                              | Discovery, not disruption; balanced on tradition axis                      |
| Santa Cruz Mtns | Obsession     | D2: +2 → +1                             | Private vision, not extreme endurance; cluster shift to Old World Interior |
| Sonoma          | Authenticity  | D1: 0 → −1, D3: 0 → −1                  | Deliberate informality; cluster shift to New World Reinvention             |
| Stellenbosch    | Aspiration    | D4: 0 → +1                              | Individual professional ambition; anchors in New World Reinvention         |
| Tuscany         | Art           | Metaphor change, D2: 0 → −1, D3: +1 → 0 | Renaissance cultural frame; balanced tradition                             |
| Wachau          | Monumentality | D1: +1 → 0, D2: +1 → 0                  | Composed grandeur, not extreme struggle; cluster shift to The Moderates    |
| Wagram          | Earth         | D1: +1 → +2, D3: +1 → +2, D6: +2 → +1   | Deeply earthly; strong custodial tradition of loess landscape              |
| Walla Walla     | Community     | D1: 0 → +1                              | Place-anchored community identity                                          |

## D.5 Complete D-Score Matrix

All 59 regions sorted by identity cluster, then alphabetically within cluster. Columns: identity cluster, region name, country, terroir cluster (abbreviated), and the six D-scores. Terroir abbreviations: *Med&Volc* = Mediterranean & Volcanic, *SH&Intl* = Southern Hemisphere & International, *French* = French Viticultural, *Amer.W* = American West Coast, *Iber.* = Iberian Continental, *Germ.* = Germanic Rhine, *Austr.* = Austrian Danube.

| Cluster                      | Region                | Country     | Terroir  | D1 | D2 | D3 | D4 | D5 | D6 |
|------------------------------|-----------------------|-------------|----------|----|----|----|----|----|----|
| <i>The Moderates</i>         | Alsace                | France      | French   | 0  | 0  | +1 | 0  | 0  | 0  |
|                              | Alto Adige            | Italy       | SH&Intl  | 0  | 0  | +1 | 0  | 0  | +1 |
|                              | Friuli-Venezia Giulia | Italy       | Med&Volc | -1 | 0  | 0  | 0  | 0  | 0  |
|                              | Goriška Brda          | Slovenia    | Med&Volc | -1 | 0  | 0  | 0  | 0  | +1 |
|                              | Hawke's Bay           | New Zealand | SH&Intl  | 0  | 0  | 0  | 0  | -1 | +1 |
|                              | Kamptal               | Austria     | Austr.   | +1 | 0  | +1 | +1 | 0  | 0  |
|                              | Loire                 | France      | French   | -1 | 0  | 0  | 0  | 0  | 0  |
|                              | Maipo Valley          | Chile       | SH&Intl  | -1 | 0  | +1 | 0  | -1 | +1 |
|                              | Margaret River        | Australia   | SH&Intl  | 0  | 0  | 0  | 0  | -1 | 0  |
|                              | Nahe                  | Germany     | Germ.    | +1 | 0  | +1 | +1 | -1 | 0  |
|                              | Santa Barbara         | USA         | Amer.W   | -1 | 0  | 0  | 0  | 0  | +1 |
|                              | Steiermark            | Austria     | Med&Volc | +1 | 0  | 0  | +1 | +1 | +1 |
|                              | Tuscany               | Italy       | Med&Volc | -1 | -1 | 0  | +1 | 0  | 0  |
|                              | Wachau                | Austria     | Austr.   | 0  | 0  | +1 | +1 | 0  | +1 |
|                              | Willamette Valley     | USA         | Amer.W   | -1 | +1 | -1 | +1 | 0  | -1 |
| <i>New World Reinvention</i> | Catalonia             | Spain       | Iber.    | -1 | 0  | -1 | +1 | +2 | +1 |
|                              | Central Otago         | New Zealand | SH&Intl  | -1 | +1 | -2 | +1 | +1 | +1 |
|                              | Columbia Valley       | USA         | SH&Intl  | -1 | +1 | -1 | 0  | +1 | +2 |
|                              | Finger Lakes          | USA         | SH&Intl  | -1 | +1 | -1 | +1 | +1 | +1 |
|                              | Mendoza               | Argentina   | Iber.    | -2 | 0  | -2 | +1 | +2 | +1 |
|                              | Napa Valley           | USA         | Amer.W   | -2 | 0  | -1 | +1 | +1 | +1 |
|                              | Paso Robles           | USA         | Amer.W   | -1 | 0  | -1 | +2 | +1 | +1 |
|                              | Patagonia             | Argentina   | SH&Intl  | -1 | +2 | -2 | +1 | +1 | +1 |
|                              | Sicily                | Italy       | Med&Volc | -1 | +1 | -2 | +1 | +1 | +1 |
|                              | Sonoma                | USA         | Amer.W   | -1 | 0  | -1 | +1 | 0  | +1 |
|                              | Stellenbosch          | S. Africa   | SH&Intl  | -1 | 0  | -1 | +1 | +1 | +1 |
|                              | Swartland             | S. Africa   | Med&Volc | -1 | +1 | -2 | +1 | +2 | +1 |
| <i>Old World Interior</i>    | Burgundy              | France      | French   | +2 | +1 | +2 | +1 | -1 | -1 |
|                              | Campania              | Italy       | Med&Volc | +1 | 0  | +1 | 0  | -1 | -1 |
|                              | Etna                  | Italy       | Med&Volc | +1 | +2 | 0  | 0  | +1 | -1 |
|                              | Galicja               | Spain       | Iber.    | +1 | +1 | +1 | +1 | 0  | -1 |
|                              | Mosel                 | Germany     | Germ.    | +2 | +2 | +2 | +1 | -2 | -2 |
|                              | Northern Rhône        | France      | French   | +2 | +2 | +2 | +2 | -2 | -1 |
|                              | Piedmont              | Italy       | Med&Volc | +2 | +1 | +2 | +1 | -2 | -1 |
|                              | Rheingau              | Germany     | Germ.    | +1 | 0  | +2 | +1 | -2 | 0  |
|                              | Santa Cruz Mountains  | USA         | Amer.W   | +2 | +1 | 0  | +2 | 0  | 0  |
|                              | Tokaj                 | Hungary     | Med&Volc | +2 | +1 | +2 | +1 | -2 | -1 |
| <i>Against the Odds</i>      | Barossa Valley        | Australia   | SH&Intl  | +1 | +2 | +1 | 0  | 0  | +1 |
|                              | Douro                 | Portugal    | Iber.    | +1 | +2 | +2 | +1 | -1 | +1 |
|                              | Hunter Valley         | Australia   | SH&Intl  | +1 | +2 | +1 | +1 | -1 | +1 |
|                              | Jura                  | France      | French   | +2 | +1 | +2 | +1 | 0  | +1 |
|                              | Liguria               | Italy       | Med&Volc | +2 | +1 | +1 | +1 | -1 | +1 |
|                              | Macedonia             | Greece      | Med&Volc | +2 | +2 | +2 | +1 | 0  | +1 |
|                              | Ribera del Duero      | Spain       | Iber.    | +1 | +2 | +1 | +1 | 0  | +2 |
|                              | Santorini             | Greece      | Med&Volc | +2 | +2 | +1 | +1 | +1 | +1 |
|                              | Sardinia              | Italy       | Med&Volc | +2 | +1 | +2 | -1 | -1 | +2 |
|                              | Wagram                | Austria     | Austr.   | +2 | 0  | +2 | 0  | 0  | +1 |
| <i>Old World Exterior</i>    | Baden                 | Germany     | Germ.    | -1 | -1 | +1 | -1 | -1 | +1 |
|                              | Bordeaux              | France      | French   | -2 | -1 | +2 | -2 | 0  | +1 |

| Cluster             | Region              | Country     | Terroir  | D1 | D2 | D3 | D4 | D5 | D6 |
|---------------------|---------------------|-------------|----------|----|----|----|----|----|----|
| <i>Outward Ease</i> | Champagne           | France      | French   | -2 | -1 | +2 | -2 | 0  | -1 |
|                     | Châteauneuf-du-Pape | France      | French   | -1 | 0  | +2 | -2 | -1 | +1 |
|                     | Dalmatian Coast     | Croatia     | Med&Volc | +2 | -2 | +1 | -1 | -2 | +2 |
|                     | Pfalz               | Germany     | Germ.    | -1 | -2 | +1 | -1 | -1 | +1 |
|                     | Rioja               | Spain       | Iber.    | -1 | 0  | +2 | -1 | -2 | 0  |
|                     | Beaujolais          | France      | French   | -1 | -1 | -1 | -1 | +1 | +1 |
|                     | Marlborough         | New Zealand | SH&Intl  | -2 | 0  | -1 | -1 | +2 | +1 |
|                     | Provence            | France      | French   | -2 | -2 | 0  | -1 | +1 | +1 |
|                     | Veneto              | Italy       | Med&Volc | -2 | 0  | -1 | -1 | +1 | +1 |
|                     | Walla Walla         | USA         | Amer.W   | +1 | -1 | -1 | -2 | +1 | +1 |

## D.6 Pipeline Parameters

A consolidated reference of all parameters used in the analytical pipeline.

| Parameter                         | Value                                             |
|-----------------------------------|---------------------------------------------------|
| TF-IDF <code>ngram_range</code>   | (1, 2) — unigrams and bigrams                     |
| TF-IDF <code>max_features</code>  | 800                                               |
| TF-IDF <code>sublinear_tf</code>  | True (logarithmic term frequency)                 |
| Standardisation                   | <b>StandardScaler</b> on each layer independently |
| PCA components (identity)         | 6 (100% variance retained)                        |
| PCA components (terroir)          | 10 (32.2% variance retained)                      |
| K-means $K$ (identity)            | 6, selected by silhouette maximisation            |
| K-means $K$ (terroir)             | 7, selected by silhouette maximisation            |
| K-means <code>n_init</code>       | 20 random restarts                                |
| K-means <code>random_state</code> | 42                                                |
| $k$ -NN graph $k$ (both layers)   | 5                                                 |
| Symmetrisation                    | Union of mutual top-5 neighbour sets              |

## D.7 Independence Test Results

The two clusterings, evaluated for agreement.

| Statistic                          | Value |
|------------------------------------|-------|
| Identity silhouette                | 0.303 |
| Terroir silhouette                 | 0.246 |
| Adjusted Rand Index                | 0.023 |
| Chi-squared statistic              | 38.78 |
| Degrees of freedom                 | 30    |
| Chi-squared $p$ -value             | 0.131 |
| Permutation-test percentile of ARI | 47th  |

## D.8 Pipeline Configuration Comparison

Three pipeline configurations were tested; the chosen design is Stage 3 (D-scores only).

| Stage | Inputs                                       | Silhouette ( $K = 6$ ) | Selected |
|-------|----------------------------------------------|------------------------|----------|
| 1     | Layer 1 TF-IDF only                          | 0.114                  |          |
| 2     | Layer 1 TF-IDF + D-scores ( $\alpha = 0.7$ ) | 0.117                  |          |
| 3     | D-scores only                                | 0.298                  | ✓        |

## D.9 Source Data Files

The analytical results are shipped with the Region Affinities web tool as static JSON files in the `public/data/` directory. The files and their contents are:

| File                               | Contents                                                                                                                           |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <code>regions.json</code>          | 59 regions with country, world, metaphor, identity cluster, and D-scores; metadata block with cluster summaries                    |
| <code>similarities.json</code>     | Full $59 \times 59$ identity and terroir cosine similarity matrices, top-5 neighbour lists, and per-region disagreement statistics |
| <code>identity_pca.json</code>     | Region coordinates in the first two identity PCs                                                                                   |
| <code>terroir_pca.json</code>      | Region coordinates in the first two terroir PCs                                                                                    |
| <code>terroir_clusters.json</code> | Region-to-terroir-cluster mapping                                                                                                  |

For the source narratives that drive the identity layer, see *Region Reference — Identity*. For the structured terroir profiles that drive the terroir layer, see *Region Reference — Terroir*.

## D.10 Reproducibility

The analytical pipeline is deterministic given the random seed (`random_state = 42`). Software versions: Python 3.11, scikit-learn 1.4, scipy 1.12, numpy 1.26, pandas 2.1. The pipeline can be rerun from the source profiles to reproduce all numerical results in this document exactly.